Proposal for the Fish and Wildlife Compensation Program

Design and Work Plans for Gravel Placement in Jordan River Reach 1

Project Proposal

Prepared for:

FWCP Program Manager

BC Hydro

6911 Southpoint Drive – E16

Burnaby, BC, V3N 4X8

November 6, 2015

Prepared by:

Helen Jones, Pacheedaht First Nation, Port Renfrew, BC

David Burt, D. Burt and Associates, Nanaimo, BC

Graham Hill, Northwest Hydraulic Consultants, Nanaimo, BC

PROJECT PROPOSAL

1. Introduction

The project site for this proposal is located in west side of the Jordan River about 80 m downstream of the Jordan River Generating Station tailrace (**Figure 1**). This location has been identified as Prescription Site 5 in the Jordan River Restoration Plan (Burt and Hill 2015). The ultimate plan for this prescription site involves the installation of spawning gravel, however, the goal of this current project is to complete any needed data collection, analyses, and reporting to allow this project to move to the construction phase via a FWCP submission in November 2016 for construction in 2017. In particular, this project will develop spawning platform design, engineering drawings, work plan, costing, hydrology/hydraulic/stability analyses, and others.

An understanding of the tidal influences, salinity regimes, water level/discharge and substrate size/stability relationships will be supported by another project funded by BC Hydro's Jordan River Seismic Project and being conducted on the river from approximately December 2015 to May 2016. This project involves a field and office program for development of a 2-dimenstion hydraulic model of Reaches 1 and 2 of the Jordan River. Intergravel salinity monitoring is a part of this study. Once complete, the model can be used to assess and compare hydraulic conditions at Prescription Site 5 under various discharge scenarios, as well as assess any effects of restoration prescriptions on tailrace operation.

2. Background

The Jordan River is located adjacent to the village of Jordan River about 30 km west of the town of Sooke. The portion of the river accessible to anadromous salmonids is only 1,625 m long and consists of three reaches (Figure 1). Reach 1 is tidal, 970 m long, and extends from just below the highway bridge to 80 m upstream of the Jordan River Generating Station tailrace. Reach 2 is 330 m in length and extends from the top of Reach 1 upstream to a 3.0 m falls. Reach 3 extends from this falls for 325 m to a second falls 4.6 m in height. This latter falls is considered to be the upper limit for anadromous fish.

The area of focus for this proposal is a 65 m long section of river on the right side of Reach 1 with its upstream end located about 80 m downstream of the Jordan River tailrace (Figure 1). This site currently has some existing gravel but the quality is diminished by the shallowness of the layer, interspersing of small and large cobbles, and by the presence of 15% fines. The intent of this project is to provide the information and documentation to allow this project to move forward with Phase I of the Site 5 plan in a future FWCP submission.

Historically, the Jordan River supported runs of about 700 coho, 1500 chum, and 3000 pinks. The river was also known to contain runs of steelhead and anadromous cutthroat trout. These runs were virtually eliminated beginning in the 1950's (coho and chum), and 1970's (pink), probably due to the effects of copper toxicity, compounded by construction of the dryland sort in the estuary, and by movement of BC Hydro's generating station/tailrace facility from the east side of the river to the west side (Burt 2012). The consequences of high levels of copper in the river were demonstrated by

juvenile fish population assessments which found a total absence of any rearing fish until progressing upstream of the copper affected zone (Burt 2013, Griffith 1996).

Two recent events have dramatically turned around the effects of copper on rearing stages of salmonids. The first was several very large floods during the winter of 2006/07, which appeared to scour some of the source material from the river. The second, and most important, was the dilution factor provided by fish flow releases initiated from Elliott Dam in January 2008. Beginning in the summer of 2008, various age classes of rainbow trout, as well as coho fry have been found rearing in the copper affected zone of Reach 2. Furthermore, these fish were found to have weight to length ratios (condition factor) similar to fish upstream of the copper affected zone. These results suggest that the dilution factor is sufficient that rearing stages are now able to feed and grow as normal (Burt 2012, 2013). Another positive sign is that small number of adult pink, coho, chum, and steelhead have been observed in Reaches 1 and 2 in recent years (Cascadia Biological Services 2013; Dwayne Walsh, Jordan River Generating Station, pers. comm.).

Despite the above improvements in water quality, recovery of salmonids to the river remains a challenge due to shortages of both rearing and spawning habitat within the anadromous river length. This proposal represents a start at addressing the shortage of spawning habitat. Shortages of spawning habitat are particularly relevant to species such as pink and chum salmon. These species have minimal freshwater rearing and thus freshwater production tends to be dictated by available spawning habitat. Interview of T'Sou-ke Nation Elders (Recreation Resources Ltd 2001) suggested that historically Reach 1 was heavily used for spawning by pink and chum salmon (tidal spawners), while coho and steelhead relied on reaches 2 and 3 for spawning.

3. Project Objectives and Approach

In 2015/16, BC Hydro, in coordination with Pacheedaht First Nation, Fisheries and Oceans Canada, BC Ministry of Forests Lands and Natural Resource Operations, D. Burt and Associates, and Northwest Hydraulic Consultants Ltd (NHC), are undertaking an important project to collect detailed hydrometric and survey data of the river, and develop a 2-D modelling tool that will be used for future Jordan River Restoration work. This FWCP submission intends to use the 2-D Model Project for its intended purpose of characterizing baseline fish habitat and river conditions, and then leveraging the data and the model to developing detailed fish habitat and restoration prescription designs for Site 5.

There are three main objectives of the proposed work:

1. Conduct field studies and analysis to confirm the viability of gravel placement at the proposed site in Reach 1 of the Jordan River.
2. Establish and maintain a hydrometric, salinity, temperature, and DO monitoring station at the proposed site.
3. Prepare a project design, work plan, and all necessary documentation to allow this project to be constructed in a future FWCP submission.

The ultimate goal of the project is to install high quality spawning gravel at the proposed site with the target species being pink and chum salmon (known tidal spawners). For the 2016/17 period (this grant proposal) the background studies, data collection, planning, and design work will be

carried out. These objectives are consistent with the first two “Priority Topics” outlined in Section 4.3 of the Jordan Watershed Plan (BC Hydro 2011), namely 1) re-introduction of pink salmon, and 2) multiple fish species habitat restoration.

It is also the goal of this project to involve local land owners, stake holders, First Nations, and agencies to garner support for project objectives and to encourage participation in future restoration initiatives. Indeed, recent dialogue by the Pacheedaht with local sources for machinery and gravel have demonstrated enthusiastic support for restoration of the Jordan River and for participating in restoration efforts.

The approach proposed to achieve the study objectives involves the following field and office tasks:

Field Tasks

- A field visit to develop the site plans, and identify machine access for a test pit program.
- The test pit program includes digging test holes to assess subsurface conditions and check for the presence of bedrock. The pits will be filled in to match the existing grades.
- Water level, temperature, DO, and salinity sensors will be pre-installed by the 2-D Model Project, however, that project will end at the time this one begins, and so a part of this projects tasks will be to continue with the download, workup, and analyses of the data from these loggers.
- River currents at the project site will be measured with an Acoustic Doppler Current Profiler (ADCP).
- Snorkel surveys to assess adult salmon usage within the Jordan River (including existing gravel at the project site).
- Provide a site tour for stakeholders and interested parties including First Nations, agencies, property owners, and local newspaper representative (e.g., Sooke Mirror).

Office Tasks

- Review the hydrologic analysis (2-D Model Project), and adapt it for this work.
- Run 2-D model to assess hydraulic conditions at Site 5 as a function of flow; compare with- and without-project conditions (output from 2-D Model Project).
- Conduct a hydraulic forces analysis and determine gravel sizes and their stability.
- Describe intergravel temperature and salinity regimes and relate to egg and alevin tolerances (output from data collected by 2-D Model Project and biological interpretation).
- Develop detailed design specifications (construction drawings with site location, plan, profile, and cross-sections, access route, and laydown area).
- Develop a work plan (construction equipment and methods, source of materials, schedule, and costing of materials and construction).
- Identify and map land ownership.
- Solicit letters of support from First Nations, agencies, and property owners.

- Form strategic partnerships, solicit in-kind contributions, and identify partnership funding opportunities.
- Prepare environmental and safety plans for the construction project.
- Compile the above into a project report.

4. Methods

Field Investigations

Much of the field investigations that would normally be required to provide baseline data for this project will be collected by the 2-D Model Project, (BCH, 2015/16). The purpose of remaining field investigations are to develop project concepts and further advance both the physical engineering and biological components of the design. These investigations will include digging test pits, conducting snorkel surveys, monitoring sensors, measuring velocities and flow directions at the gravel site, and providing a site tour to stakeholders.

The purpose of the test pitting is to investigate the subsurface soils at the proposed gravel platform site. It is important to assess the subsurface conditions, including determining if marine sediments or bedrock are near the surface. Up to three test pits will be excavated using a 20 tonne hydraulic excavator, such as a Hitachi EX200, during a low flow period in May 2016. The test pits will be excavated to a depth of 1 m below the existing stream bed. It is important to note that the test pits will only characterize the subsurface conditions of its footprint; conditions laterally and longitudinally may vary.

Snorkel surveys will be undertaken within the anadromous portion of the Jordan River in order to update existing knowledge on adult salmon returns (species, abundance, timing, and areas of utilization). This will include monitoring for potential usage of existing gravel at the proposed project site. Snorkel surveys will be performed by the Pacheedaht FN fisheries crew. Four snorkel surveys will be undertaken spanning the period of late September to late November.

Hydraulic parameters such as flow magnitude and direction will be measured at the project site using an ADCP instrument. This data will be used to refine and verify the model at the project site, as well as provide direct design input into the proposed spawning gravel project.

Hydrologic Analysis

A high flow and low flow frequency and exceedance analysis will be completed as part of the 2-D Model Project (BCH, 2015/16). We will re-familiarize ourselves with this data and adapt it to the design task for this project. The 2-D Model Project hydrology work includes:

1. A hydrologic assessment for the lower river below the tailrace, this is important for designing effective restoration projects. The key hydrologic factors for Reach 1 are tides, tailrace discharge, and upper river discharge.
2. Regional tidal statistics will be adjusted from chart datum to geodetic datum, and related to the site.
3. There are three main discharge data sets. Each set has a period of record from 1990 to 2013. The data summary is as follows:
    a. Daily discharge into Elliot Headpond.

   b. Daily and instantaneous discharge from Elliot Headpond.
   c. Daily and instantaneous discharge from the generating station.
4. Discharge statistics in the anadromous reach are separated into two sections. The upper section is upstream of the generating station, and the lower section is downstream of the generating station. For the upper section the discharges are estimated by:
   a. Establishing a regional daily discharge to instantaneous discharge relationship based on analysis of nearby gauges.
   b. Developing a synthesised instantaneous discharge record into Elliot Headpond based on the regional relationship from (1).
   c. Estimating the daily and instantaneous discharge records for the watershed area downstream of Elliot Headpond and above the generating station based on an area relationship and the data records from (2). Elevation and precipitation data, if available may be factored in.
   d. Combining daily and instantaneous discharges from (1) and (3).
5. Discharge records for the lower reach are developed by combining (4) with the discharge records from the generating station.
6. A high flow frequency analysis was carried out using the discharges records. Daily and instantaneous discharges will be reported for various return intervals up to the 50-year flood. The low flow analysis includes 7-day low flow extremes.

These hydrologic inputs will be used to model and assess the expected performance of the proposed gravel project.

Hydraulic analysis

The approach for this study is to estimate the important hydraulic parameters required for the design of the pilot gravel platform. The assessment will include estimates of the minimum and maximum depths, flow velocity in relation to river discharge, an estimate of shear stress, and an analysis of the stable particle size for the spawning gravel.

A 2-dimensional flow and fish habitat model is the most comprehensive tool for assessing the reach hydraulics. The model developed from the 2-D Modeling Project will be adapted for this work task.

Another use of the flow model is to assess hydraulic changes for with- and without the project. This analysis will be used to describe any expected water level or flow path changes in the river. Past FWCP applications comments have queried whether the gravel projects will cause any changes to the generating station or other BCH operations. The model will be used to help assess and quantify whether any changes are expected.

Spawning Platform Design

The design will include a set of engineered plans and specifications for constructing the spawning platform. The drawing set will include:

1. A project location map.
2. Identification of the site access route and staging area.
3. Plan, profile, and cross-sections of the project with suitable detail for a contractor experienced with river and fish habitat work to construction the project.

4. Materials specification including the spawning gravel, access route surfacing, disturbed area remediation and plant list for revegetation.

5. Deliverables

The deliverable from this project will be a report that includes the spawning platform design drawing, an installation work plan, results of the hydrologic and hydraulic analysis, results of the environmental monitoring (intergravel salinity, temperature, and DO).

6. Value-Added Criteria

Activities to achieve value-added benefits will include the following:

1. Establishment of a Jordan River Round Table— There are already the beginnings of a Jordan River Round Table comprised of First Nations, local stake holders, land owners, and agencies. This project will build on the developing these relationships, increasing awareness of the FWCP program, and garnering support for future restoration works on the Jordan River. Ultimately, the plan is for the Round Table to oversee restoration efforts on the river.
2. The Jordan River has long been considered a “dead river” due to impacts of copper inputs at the top of Reach 2. This situation changed dramatically in 2008 with the onset of fish flow releases from Elliott Dam. These have provided sufficient dilution of the copper that the previously dead zone is now utilized by rearing and overwintering coho, rainbow, and possibly steelhead (these are the stages most sensitive to copper). This project will help to convey current knowledge and status of the Jordan River, and when combined with restoration efforts, there is potential for the Jordan to demonstrate a tremendous success story in the recovery of historic salmon runs.

7. Risks and Benefits

No risks of negative impacts are anticipated in this year’s funding submission since the proposed work involves field surveys and design development. Risks of potential negative impacts associated future gravel installation at the project site will be incorporated into the report generated by this project.

Similarly, biological benefits of this project will not be realized until it moves to the construction phase (anticipated to be 2017). These will be identified in the report as well as potential benefits to non-target environmental components.

The main benefit from the works outlined in this proposal is that it provides a venue to pull together the various user groups and stake holders, to disseminate current knowledge on the status of the river, and to involve them in the restoration process.

8. Performance Measures

Monitoring of work progress and consistency with study objectives will be achieved by drafting a project timeline with achievement milestones. Monthly budget and progress reports will be completed and compared with project milestones.

Monitoring and evaluation of the success of spawning gravel prescription will be part of future proposals when the actual works has been completed. If results from this proposal demonstrate the project is viable, then a submission for funding to install the platform would occur in the November 2016 FWCP intake.

9. Past Projects

List of previous relevant projects implemented by the study team.

Contact Information	Project Description
D. Burt and Associates Project Contact: Dave Ewart (DFO) 250-287-9564 Mike Gage (Campbell River Salmon Foundation) 250-287-4368	Reviewed the success and biological utilisation of existing restoration works on the lower Campbell River and developed conceptual plans for future restoration efforts. Report: "Restoration of the Lower Campbell River, A Review of Projects to 2003 and a Plan for Future Works" (Burt 2004).
NHC Project Contact: Martin Buchanan (Campbell River Salmon Foundation) 250-923-8507 Mike Gage (Campbell River Salmon Foundation) 250-287-4368	Campbell River Chinook spawning gravel platform design, construction and monitoring program. Completed river surveys, 2-dimensional numerical modeling, and designs for three gravel sites identified by Burt (2004). Placed over 8,000 m^3 of gravel in four construction phases. (NHC 2006; DFO 2007; NHC 2008; NHC 2009; NHC 2010; NHC 2013; NHC(2) 2013)
Pacheedaht FN Project Gordon River Sidechannel 1999 Mel Sheng	In partnership with DFO, coordinated the assessment and project management of the Gordon River Sidechannel construction

10. Literature Cited

BC Hydro. 2011. Jordan Watershed, Watershed Plan, Final Draft. BC Hydro Fish and Wildlife Compensation Program, Available online at: https://www.bchydro.com/content/dam/hydro/medialib/internet/documents/about/our_commitment/fwcp/pdf/funding_donations/watershed_action_plans/small_basin_plans/jordan_watershed_plan.pdf. 22 p.

Burt, D.W. 2004. Restoration of the lower Campbell River, a review of projects to 2003 and a plan for future works. Prepared for the Campbell River Gravel Committee, the Ministry of Water, Land and Air Protection, and the Department of Fisheries and Oceans. 96 p. + map.

Burt, D.W. 2012. Summary of water quality and biological data for anadromous reaches of the Jordan River. Prepared for BC Hydro, Fish and Wildlife Compensation Program, John Hart Generating Station, Campbell river, BC. 51 p.

Burt, D.W. 2013. Lower Jordan River fish index study - year 6 results (2010) and pre/post flow release comparisons. Prepared for BC Hydro, Water Use Planning, Burnaby, BC. 59 p.

Burt, D.W., and G. Hill. 2015. A Restoration Plan for Fish and Fish Habitat within Anadromous Reaches of the Jordan River. FWCP Project 14.JOR.01 Prepared for the Fish and Wildlife Compensation Program. 65 p.

Cascadia Biological Services. 2013. Jordan River spawning assessment and enumeration, Year 6 (Sept 2010 - May 2011). WUP Monitor JORMON#3. BC Hydro, Water Use Planning, Burnaby, BC. 54 p.

Griffith, R.P. 1996. Biophysical assessment of fish production within the Jordan River Drainage, 1994. BC Hydro and Power Authority, Burnaby, BC. 129 p.

Recreation Resources Ltd. 2001. Jordan River historical salmonid survey. Prepared for BC Hydro, Burnaby, BC. 35 p.

11. Credentials and Qualifications

Key players in the study team include Dave Burt (biologist, D. Burt and Associates), Helen and Jeff Jones (technicians, Pacheedaht First Nation), and Graham Hill (Engineer, Northwest Hydraulic Consultants). Dave Burt will serve as the project lead and will be in charge of project management, as well as the main person involved in writing the report. Graham Hill will provide bio-engineering and structural stability recommendations and assist in the hydrology analysis. The following is a summary of qualifications and experience of the study team.

David Burt (D. Burt and Associates, Nanaimo, BC)

Mr. Burt is a graduate of UBC (B.Sc., 1980, honours program) and has 33 years of experience in fish habitat assessment, fish inventory, and fisheries biology. He has worked extensively in the field of assessing and mitigating the interactions between regulated streamflow and fish habitat. Mr. Burt has a track record of providing quality support to BC Hydro and agencies in a variety of fisheries studies. He is skilled in report writing, data analysis, statistics, GIS, database development, and graphics. With respect to the Jordan River, Mr. Burt has an intimate knowledge of the lower river gleaned from 8 years of study on the river and through research of its history.

Jeff Jones (Pacheedaht First Nation, Port Renfrew, BC)

Mr Jones has a Fisheries Field Technician Diploma from VIU's Fisheries and Aquaculture Extension Program. He has 20 years experience in a variety of fisheries field methods and was the primary technician in assisting David Burt in the 6 years of fisheries and hydrology studies conducted on the lower river (JOR Monitors #1 and #2). He has been part of the snorkel team

providing salmon enumerations to DFO for the San Juan and Gordon River systems for the past 10 years. Mr. Jones has also been involved in sidechannel developments in the San Juan and Gordon River systems including feasibility, implementation, assessment, and monitoring programs.

Helen Jones (Pacheedaht First Nation, Port Renfrew, BC)

Ms. Jones has a B.Sc. from the University of Manitoba, and has 20 years experience in project management and fisheries assessments. She is responsible for coordinating and implementing Pacheedaht First Nation fisheries programs. Like Mr. Jones, she has been part of the snorkel team providing salmon enumerations to DFO for the San Juan and Gordon River systems and was involved in sidechannel developments in the San Juan and Gordon River systems (feasibility, implementation, assessment, and monitoring programs).

Graham Hill (P. Eng, Northwest Hydraulic Consultants, Nanaimo, BC)

Mr. Hill has experience in project planning, analysis, design and construction supervision. He has worked in multi-disciplinary teams of engineers, biologists, technicians, planners, and environmental scientists and has experience in permitting, project implementation, and project management. His project experience includes riverbank stabilization using riprap and bio-engineering methods, fish habitat restoration including large woody debris design and spawning gravel placement, rock weir design, off channel habitat improvements, estuary and wetland restoration, dam removal mitigation, fishways, fish passage through culvert barriers, weir design, and river water intakes. He has experience in preparation of plans, specifications, and construction cost estimates. He has extensive experience in the use of field survey and monitoring equipment, river modeling, hydrology and CAD design software.

12. Communication Plan

To facilitate communication of the project, local newspapers (e.g., Sooke News Mirror) will be contacted to provide articles on the “new” biological status of the Jordan River, the objectives of this project, and of future restoration opportunities to bring salmonids back to the Jordan River. Signage at restoration sites will not be implement until future years when physical works are completed. All communications will clearly acknowledge sponsorship by the BC Hydro Fish and Wildlife Compensation Program. In addition, a PowerPoint presentation will be prepared and delivered to the Pacheedaht First Nation Chief and Council with representatives from Queesto Community Forest.

Figure 1. Orthophoto map (2013) showing the anadromous reaches of the lower Jordan River (Reaches 1, 2 and 3) and the site of the proposed future spawning platform.

PART V – SUPPORTING DOCUMENTATION

1. Letters of Support

The FWCP application requires letters of support from local land owners as well as Provincial and Federal fisheries agencies. Land owners adjacent to the project area is BC Hydro. Letters of support were obtained from the above groups and are attached to this section. The list includes:

- Fisheries and Oceans Canada (Margaret Wright, DFO, Nanaimo)
- Ministry of Environment (Mike Stalberg, MOE, Nanaimo) – forthcoming
- BC Hydro
- Pacheedaht First Nation

2. Safety and Environmental Plans

Since this project involves field work downstream of the Jordan River Generating Station, safety plans will be required prior to commencement of field activities. Safety plans have been submitted to BC Hydro for previous works on the Jordan River (including FWCP seed project 2013.JOR.01 and 2013.JOR.02) and these will be adapted for the work planned in this proposal (pending approval).